In [1]:
```
my_list = [10, 20, 30, 40, 50]
list_length = len(my_list)
print(f"The length of the list is: {list_length}")
```

```
The length of the list is: 5
```

In [2]:
```
def greet(name):
    print(f"Hello, {name}!")

greet("Alice")
greet("Bob")
```

```
Hello, Alice!
Hello, Bob!
```

In [3]:
```
def find_maximum(numbers):
    if not numbers:
        return None  # Handle empty list case
    max_value = numbers[0]
    for num in numbers:
        if num > max_value:
            max_value = num
    return max_value

my_numbers = [12, 45, 8, 27, 56]
max_result = find_maximum(my_numbers)
print(f"The maximum value is: {max_result}")
```

```
The maximum value is: 56
```

In [4]:
```
global_var = "I am global"

def demonstrate_variables():
    local_var = "I am local"
    print(f"Inside function: {local_var}")
    print(f"Inside function (global): {global_var}")

demonstrate_variables()
print(f"Outside function (global): {global_var}")
# Uncomment the line below to see the error (local_var is not defined outside the function)
# print(f"Outside function: {local_var}")
```

```
Inside function: I am local
Inside function (global): I am global
Outside function (global): I am global
```

In [5]:
```
def calculate_area(length, width=5):
    return length * width

# Example usage:
length1 = 10
width1 = 3
area1 = calculate_area(length1, width1)
print(f"Area with custom width: {area1}")

length2 = 8
area2 = calculate_area(length2)  # Uses default width of 5
print(f"Area with default width: {area2}")
```

```
Area with custom width: 30
Area with default width: 40
```

In [ ]: